

Quick Check

Lesson 2: What Is Cell Division? Meaning of Cell Division and the Structure of the Chromosome

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.4: Cell Division | Lesson 2

A group compares printed pictures of a cell before and after division. Before division they count 1 cell, and after division they count 2 cells. Which conclusion is best supported by this evidence?

- A** Food that we eat turns directly into new cells
- B** One existing cell divides to make new cells
- C** A cell only gets bigger and bigger without making new cells
- D** New cells are brought into the body from outside, one by one

In a growing child, one cell divides into 2 cells. Then each of those 2 cells divides once. How many cells are there after the second round of division?

A 6

B 8

C 3

D 4

A group builds a clay model of one cell dividing into two cells. They put a nucleus in the original cell. What must the two new cells receive so that they can work like the original cell?

- A** Only extra food, because the new cells have no need for anything else
- B** A copy of the genetic information (instructions) found in the nucleus
- C** Nothing, because each new cell makes its own instructions from scratch
- D** Only a bigger cell size than the original cell

Which statement best gives the meaning of cell division?

- A** A parent cell dividing to form new daughter cells
- B** A cell becoming bigger without changing in number
- C** Two chromosomes joining to form one cell
- D** Food being changed into new cells

A chromosome has two identical arms that are joined together at one point. What is the name and job of the part where the arms are joined?

- A** A small separate cell that makes the chromosome
- B** The cell membrane, which keeps the chromosome inside the cell
- C** The gene, which is a second chromosome inside the first
- D** The centromere, which holds the two chromatids together

A chromosome is made of two identical arms joined at the centromere. What are these two identical arms called?

- A** Two separate cells
- B** Two different chromosomes
- C** Two sister chromatids
- D** Two centromeres

Humans have 46 chromosomes in their cells and maize has 20. Which statement about chromosomes and DNA is correct?

- A** Chromosomes are made of cells, and DNA is a type of chromosome
- B** Chromosomes and DNA are two completely unrelated parts of a cell
- C** Chromosomes contain DNA, and genes on the DNA carry the genetic information a new cell needs
- D** Only humans have chromosomes, and maize has only DNA

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).